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A Idk Dataset Construction Details

A.1 Data Statistics

We use the training set of TriviaQA, consisting of 87,622 samples, to construct the training and development sets of the Idk dataset. We partition 10% of the training set of TriviaQA to serve as the validation set of the Idk dataset, with the other 90% as the training set. Therefore, the validation set contains 8,763 samples and the training set contains 78,899 samples. We use the development set of TriviaQA to construct the test set for the Idk dataset, which comprises a total of 11,313 samples. The number of samples in each part of the Idk dataset for different models is the same, it is only the distribution of the labels that varies.

A.2 Sampling Parameters

When constructing the Idk dataset through sampling model responses, our sampling parameters are set as follows: `top_p=0.9`, `temperature=1.0`, `max_new_tokens=512`, `repetition_penalty=1.0` (no penalty). We use this set of parameters for all random sampling in this work.

A.3 Label Distribution of Idk Datasets

In Figure 5 and Figure 6, we present the label distribution in the Idk datasets constructed using different Ik thresholds across various models. It is evident that different models possess varying knowledge reserves, as indicated by the distinct differences in the label distribution of their Idk datasets. As shown in Figure 6, the larger the size of the model, the more extensive its knowledge, resulting in fewer questions being labeled as “I don’t know”.

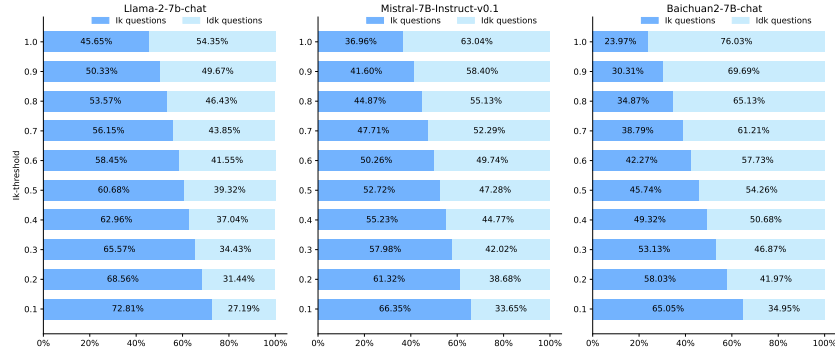


Figure 5: Label distribution in the Idk dataset across different models.

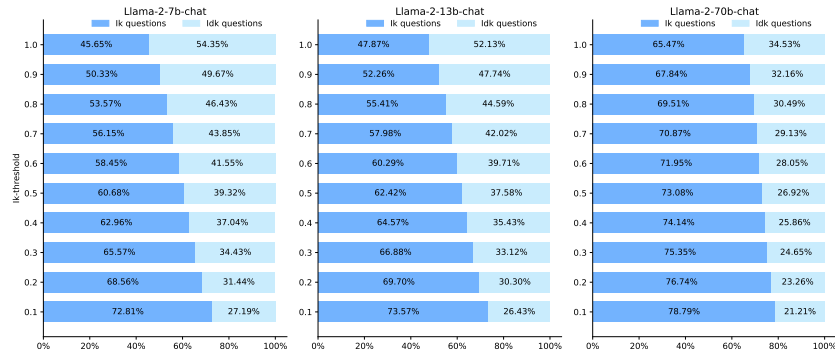


Figure 6: Label distribution in the Idk dataset across different sizes.

B Training Details

B.1 Supervised Fine-tuning

We organize our Idk dataset into single-turn dialogues following the conversation format of Llama-2-7b-chat and then use the standard SFT loss to train the model:

$$\mathcal{L}_{SFT} = -E_{(x,y) \sim D} \left[\frac{1}{N} \sum_t \log p(y_t | x, y_{<t}; \theta) \right] \quad (1)$$

(x, y) is a question-answering pair in the Idk dataset, where x represents the question, and y represents the answer. N represents the length of the answer y , and θ represents the model parameters. During training, we employ a packing strategy to combine multiple samples into a single sequence with a maximum length of 4096. Following the settings of llama-recipes⁴, our batch size is set to 32, with a learning rate of 1e-4 and train 10 epochs. During training, we save a checkpoint at the end of each epoch, and select the checkpoint that performs the best on the validation set as the final model. We employed Fully Sharded Data Parallelism (FSDP) to conduct SFT training on eight A100 80G GPUs. For Llama-2-70b-chat, we train 10 epochs using 32 A100 80G GPUs and select the checkpoint of the last epoch as the final model. The decision to forego the use of a validation set for model selection was based on our observation that the model exhibiting the lowest loss on the validation set tended to erroneously reject numerous Ik questions. We speculate that this may be attributed to the inherent alignment training of the Llama-2-70b-chat itself.

B.2 Direct Preference Optimization

The original DPO loss proposed by Rafailov et al. (2023) is:

$$\mathcal{L}_{DPO} = -E_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]. \quad (2)$$

where π_{ref} is the SFT model trained with half of the Idk data, π_θ is the policy model, y_w is the chosen response and y_l if the rejected response. To alleviate the problem of the DPO model sometimes failing to fully generate the Idk template, we additionally incorporate the SFT loss. Our final loss function of direct preference optimization is:

$$\mathcal{L}_{DPO-SFT} = \mathcal{L}_{DPO} + \alpha * \mathcal{L}_{SFT} \quad (3)$$

In the experiment, we set the coefficient α of the SFT loss to 0.01. The hyperparameters of our SFT model training are the same as Appendix B.1. During DPO training, following DPO’s official implementation⁵, we set our batch size to 64, the learning rate to 5e-7, β to 0.1 and train for one epoch. We partition 10% of the preference data to construct a validation set to select the best checkpoint. We use 8 A100 80G GPUs for DPO training. We present the impact of different α values on the model’s TRUTHFUL rate in Table 3.

Table 3: The impact of the coefficient α of the SFT loss on the model’s TRUTHFUL rate.

	$\alpha = 0$	$\alpha = 0.01$	$\alpha = 0.1$	$\alpha = 0.5$	$\alpha = 1.0$
Ik-threshold=0.5	74.28	72.39	72.06	72.31	72.08
Ik-threshold=1.0	66.14	77.89	76.68	75.55	75.72

As shown in Table 4, when using the Idk dataset constructed with Ik-threshold=0.5 for DPO training, the model is capable of accurately generating the Idk template. In this scenario, incorporating SFT loss reduces the model’s TRUTHFUL rate. However, when using the Idk dataset constructed with Ik-threshold=1.0 for DPO training, the model occasionally fails to accurately generate the Idk template. In such cases, employing a coefficient of 0.01 yields the most effective mitigation.

⁴<https://github.com/facebookresearch/llama-recipes>

⁵<https://github.com/eric-mitchell/direct-preference-optimization>